

Inference—How Close? How Con dent?

[Code ▾](#)

Foundational Concepts

In this lecture, we're going to introduce basic concepts and principles of statistical inference, namely, how can we use information about just some members of a population to infer that information about the whole population?

The goal of statistical inference is to answer the questions “How close?” and “How con dent?” How close are the shape, center, and spread of just some of the population to the shape, center, and spread of the whole population? How con dent are we of that?

In order to make a useful conclusion, we have to make a reasonable guess about how close our random sample is to the population's values and how con dent we are of being that close. Political polls are common examples in which statistical inference is used. Suppose the reality is that 60% of voters favor Candidate A and 40% favor Candidate B. Suppose we ask 100 random future voters how they will vote in the upcoming presidential election. Some number of those will favor Candidate A. Or we can run 10 simulations in which we ask 100 voters each time, with each simulation giving us a number for Candidate A.

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```
if (!caracas::has_sympy()) {  
  caracas::install_sympy()  
}  
library(caracas)  
library(sets)  
  
A <- .60  
B <- .40
```

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```
range(stats::rnorm(100))
```

```
[1] -2.463680  2.719867
```

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```
data <- LETTERS[1:2]  
rand_sample <- mapply(\(x) sample(data, size=100, replace=TRUE), 1:10)  
rand_sample[1:10, 1:10]
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	"B"	"B"	"B"	"B"	"A"	"B"	"B"	"B"	"A"	"B"
[2,]	"A"	"B"	"B"	"B"	"A"	"B"	"B"	"A"	"B"	"A"
[3,]	"B"	"B"	"A"	"B"	"A"	"B"	"A"	"B"	"B"	"A"
[4,]	"A"	"B"	"B"	"B"	"A"	"B"	"B"	"B"	"B"	"B"
[5,]	"A"	"A"	"B"	"A"	"B"	"B"	"B"	"A"	"A"	"A"
[6,]	"B"	"A"	"A"	"A"	"B"	"A"	"A"	"B"	"B"	"B"
[7,]	"A"	"A"	"B"	"A"	"A"	"A"	"A"	"B"	"A"	"B"
[8,]	"B"	"B"	"A"	"B"	"B"	"A"	"B"	"A"	"B"	"A"
[9,]	"A"	"A"	"B"	"B"	"A"	"B"	"B"	"B"	"B"	"A"
[10,]	"A"	"B"	"B"	"A"	"B"	"A"	"A"	"A"	"B"	"A"

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```
votes <- table(rand_sample)
total <- sum(votes)
```

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```
c(votes['A']/total, votes['B']/total)
```

```
      A      B
0.498 0.502
```

In this case, the answer to “How close?” is 10%, and the answer to “How confident?” is 95%. Surprisingly, 95% of the time, a random sample of 1200 people will give us an estimate that lies within $\pm 3\%$ of the actual truth in the whole population.

We have seen two important principles:

- Randomness is involved in statistical inference.
- The conclusion of a statistical inference is an estimate, together with a pair of numbers that tells us how close to the true population value the estimate is and how confident we are that the estimate is that close.

Another kind of challenge for statistical inference is how to tell whether a coin is fairly balanced and equally likely to come up heads as tails. In practice, we put the coin “on trial.” We flip the coin 100 times and count how often it is heads and tails; we are getting random evidence about the frequency with which heads or tails arise.

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```
data <- c('T', 'H')
rand_sample <- sample(data, size=100, replace=TRUE)
rand_sample
```

```
[1] "T" "T" "H" "H" "H" "H" "H" "H" "H" "H" "T" "T" "T" "T" "H" "H" "H" "T" "T"
[19] "H" "H" "T" "H" "T" "T" "T" "T" "T" "T" "H" "H" "H" "H" "T" "H" "H" "H"
[37] "T" "T" "T" "T" "H" "T" "H" "H" "T" "T" "H" "T" "H" "T" "H" "H" "H" "T"
[55] "H" "T" "H" "H" "H" "T" "T" "H" "H" "T" "T" "H" "T" "T" "H" "T" "H" "H"
[73] "H" "T" "T" "T" "T" "T" "T" "H" "H" "T" "H" "H" "T" "H" "T" "H" "H" "H"
[91] "H" "H" "H" "H" "T" "H" "T" "T" "T" "H"
```

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```
table(rand_sample)
```

```
rand_sample  
H    T  
53  47
```

Statistical inference grues prominently in tests of medications. Suppose we want to establish that a medicine works. A new medication is given to 1000 people who have a particular ailment, and a placebo is given to another 1000 people with the same ailment. If a much higher proportion of people get well taking the new medicine than taking the placebo, then that is strong evidence that the medication has a beneficial effect. If a somewhat higher proportion of people get well taking the new medicine, then we need to do some mathematical analysis to determine how persuasive the evidence is that the medication actually has a beneficial effect. To measure the strength of the evidence, we need to understand how much variation in results we should expect from the random process of some people getting well spontaneously.

The logic of statistical inference is to compare data that we collect to expectations about what the data would be if the world were random in some particular respect. Analyses of randomness and probability allow us to quantify our condence in extrapolations from some of the data to the whole population. Randomness and probability are the cornerstones of all methods for testing hypotheses. # Concept and Applications

Concept and Applications

SAMPLING AND PROBABILITY?

tistics sharpens our knowledge of how randomness is all around us. Life is full of having to make decisions under uncertainty. Two fundamental ideas in statistics are uncertainty and variation. Probability is the foundation that helps us understand uncertainty and variation. This is why probability plays a key role in statistics. Probability is a mathematical language used to measure uncertain events. Whenever we collect data or make measurements, the process that we use is subject to variation, meaning that if the same measurements were repeated, the answer would be slightly different, due to variation.

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```
table(rand_sample)
```

```
rand_sample  
H    T  
53  47
```

Data is the raw information from which statistics are created. Data is being collected everywhere all the time. When we collect data, we convert information to numbers. Statistical thinking gives you the tools to intelligently extract information from data.

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```
rand_sample
```

```
[1] "T" "T" "H" "H" "H" "H" "H" "H" "H" "H" "T" "T" "T" "T" "H" "H" "H" "T" "T"
[19] "H" "H" "T" "H" "T" "T" "T" "T" "T" "T" "H" "H" "H" "H" "T" "H" "H" "H"
[37] "T" "T" "T" "T" "H" "T" "H" "H" "T" "T" "H" "T" "H" "T" "H" "H" "H" "T"
[55] "H" "T" "H" "H" "H" "T" "T" "H" "H" "T" "T" "H" "T" "T" "H" "T" "H" "H"
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[91] "H" "H" "H" "H" "T" "H" "T" "T" "T" "T" "H"
```

Because there is always uncertainty, a strong foundation in probabilistic reasoning gives you the basic tools needed to make sense of everything that is done in statistics. Sampling refers to selecting, at random, one or more of all possible outcomes (ideally, in proportion to their likelihoods). An example of sampling is selecting a random sample of Americans for a political poll.

Sampling helps us draw inferences about the larger population of outcomes. Drawing random samples forms the foundation of probability. Probability is all about considering the possible outcomes of an experiment and quantifying their chances.

What happens when we have real, but limited, data, for which we'd like to calculate probabilities? Suppose that you want to understand how a grocery store displays cereal boxes and whether different types of breakfast cereals are targeted to adults or children. You go into our local grocery store and notice that there are 6 rows of shelves and that each shelf has 5 boxes of cereal. You can group the 6 shelves into 3 categories: the bottom 2 shelves, the middle 2 shelves, and the top 2 shelves.

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```
cereal_df <- data.frame(row_name = c("adults", "children", "total"), top = c(9, 1, 10), middle = c(3, 7, 10), bottom = c(0, 10, 10), total = c(12, 18, 30))
cereal_df
```

row_name <chr>	top <dbl>	middle <dbl>	bottom <dbl>	total <dbl>
adults	9	3	0	12
children	1	7	10	18
total	10	10	10	30
3 rows				

What is the probability of A?

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```
P = function(E, p = cereal_df[3, "total"]) cmStats$P(E,p)
A <- cereal_df[1, "total"]
p = cereal_df[3, "total"];
```

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P(A)

```
[1] 0.4
```

We could do the same type of calculation to determine the probability of breakfast cereal being on the middle 2 rows: $B = \{\text{breakfast cereal is on the middle 2 rows}\}$.

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```
B <- cereal_df[3, "bottom"]  
P(B)
```

```
[1] 0.3333333
```

It's natural to think about how 2 events relate to each other. For example, what's the probability that the cereal is targeted at adults and is located on the middle 2 shelves? In this case, we want to look at our data to see where those 2 events occur together. Notice that 3 (not 2) types of breakfast cereal are located in the middle 2 shelves and targeted at adults. So, the probability would be

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```
P(intersect(A, B)) ~ 3/30 ~ abs(diff(union(A, B)) - 1)/p
```

```
P(intersect(A, B)) ~ 3/30 ~ abs(diff(union(A, B)) - 1)/p
```

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```
P(union(A, B)) ~ P(A) + P(B) - abs(diff(union(A, B)) - 1)/p
```

```
P(union(A, B)) ~ P(A) + P(B) - abs(diff(union(A, B)) - 1)/p
```

We call the probability of A and B the intersection, the place where the 2 events overlap.

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```
sum(P(union(A, B))) - abs(diff(union(A, B)) - 1)/p
```

```
[1] 0.6333333
```

We can update our probabilities of events by conditioning on new given information. Let's define a new event. Let C be the event that cereal is targeted to children: $C = \{\text{children breakfast cereal}\}$. What's the probability of C?

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```
P(intersection(C, B)) ~ 7 / 30
```

```
P(intersection(C, B)) ~ 7/30
```

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```
P(cereal_df[2, "middle"])
```

```
[1] 0.2333333
```

What is the probability that the cereal is targeted to children, given that it's located on the middle 2 shelves? In other words, what is the probability of event C, given the information from event B?

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```
P(cereal_df[2, "middle"], B)
```

```
[1] 0.7
```

We've limited ourselves to the middle 2 shelves, and of those 10 boxes of cereal, we see that 8 of them are targeted to children. So, the conditional probability was higher.

SAMPLE SPACES, EVENTS, AND PROBABILITIES

The building blocks of statistics are the axioms of probability. These beginning statements are the ground rules from which we build and prove statistical theory.

Probability theory can be reduced to 3 fundamental axioms.

- 1 For any event A, $0 \leq P(A) \leq 1$. In other words, the probability of any event is between 0 and 1. The smallest that a probability can ever be is 0; the largest it can ever be is 1.
- 2 If S is the sample space, $P(S) = 1$ and $P(\emptyset) = 0$. In other words, the probability of the entire sample space is 1 and the probability of the empty set is 0. The sample space is everything possible for the probability experiment. There are no events outside of the sample space.
- 3 The third axiom of probability deals with mutually exclusive events, but first we need to understand unions and intersections. A union is the probability of A or B and is denoted $P(A \cup B)$. An intersection is the probability of A and B is denoted $P(A \cap B)$. If A and B are mutually exclusive, meaning that they don't overlap, then the probability of their union is the sum of those probabilities: $P(A \cup B) = P(A) + P(B)$.

Formally, let A and B be events, with $P(B) > 0$. The conditional probability of A given that B has occurred is

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```
P(A|B) ~ P(intersect(A, B)) / P(B)
```

```
P(A | B) ~ P(intersect(A, B)) / P(B)
```

If we set these 2 equations equal to each other, we get

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```
P(A|B)*P(B) ~ P(B|A)*P(A)
```

$$P(A \mid B) * P(B) \sim P(B \mid A) * P(A)$$

At first, this might not seem very useful. In fact, it seems rather circular that we've just rewritten $P(A \text{ and } B)$ and $P(B \text{ and } A)$ and set them equal. But if you divide both sides by $P(B)$, you're left with a famous and useful result that relates conditional probabilities known as Bayes's rule.

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$$P(A \mid B) \sim P(B \mid A) * P(A) / P(B)$$

$$P(A \mid B) \sim P(B \mid A) * P(A) / P(B)$$

Therefore, another form of Bayes's rule is

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$$P(A \mid B) \sim P(B \mid A) * P(A) / P(B) \sim P(B \mid A) * P(A) / (P(B \mid A) * P(A) + P(B \mid A^c) * P(A^c))$$

$$P(A \mid B) \sim P(B \mid A) * P(A) / P(B) \sim P(B \mid A) * P(A) / (P(B \mid A) * P(A) + P(B \mid A^c) * P(A^c))$$